Name : Yash Chaudhari

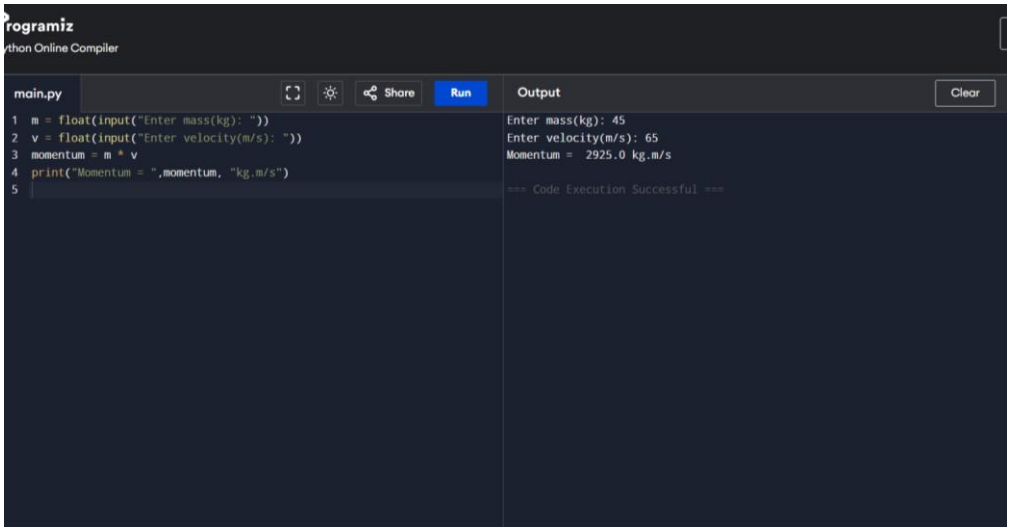
PRN : 202501080014

Branch : IT1(I11)

Practice Lab Assignment : 1

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum. Momentum is calculated as momentum= mass*velocity where m is the mass of the object and v is its velocity.

Code :



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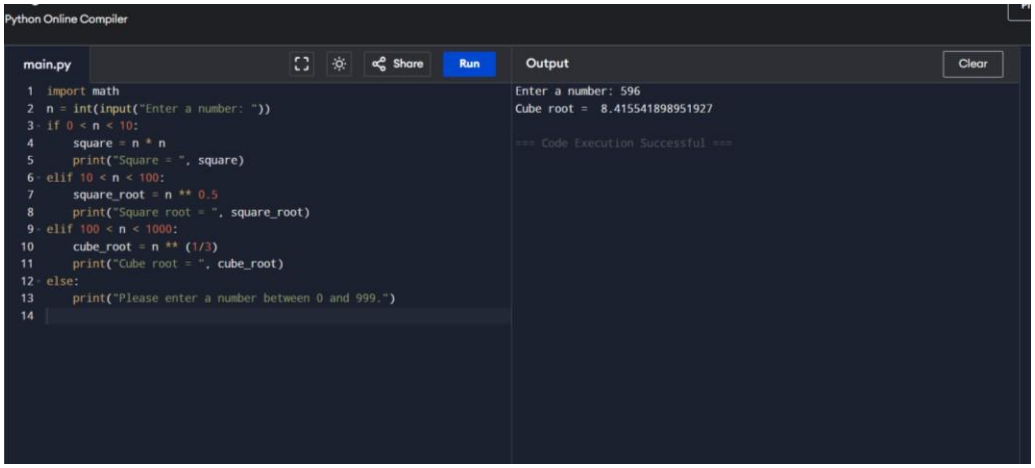
2. Write a Python program for following conditions.

- If n is single digit print square of it.
- If n is two digit print square root of it.
- If n is three digit print cube root of it.

Code :



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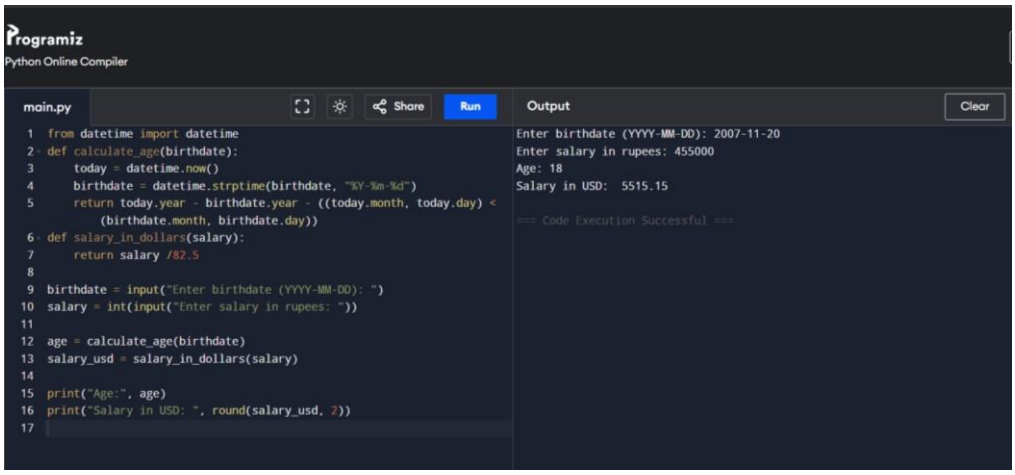
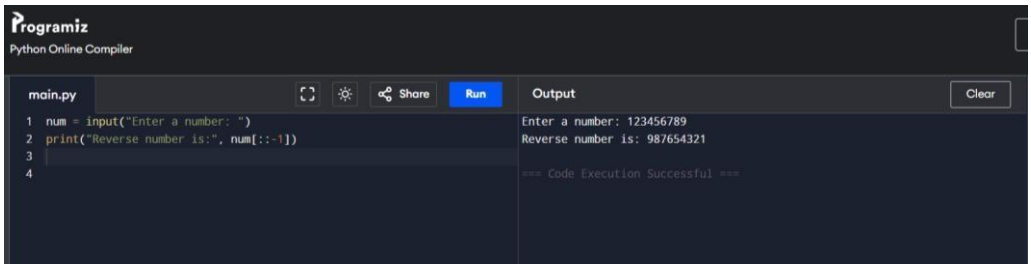
3. Read the birthdate and salary in rupees of employees. Perform data transformation for birthdate to age and also salary which is in rupees to salary in dollars using functions.

Code :



4. Print the reverse number of a given number.

Code :



Lab Assignment: 1

1. To accept students five courses marks and compute his/her result. Student is passing if he/she scores marks equal to and above 40 in each course. If student scores aggregate greater than 75 percentage, then the grade is distinction.

- If aggregate is greater than or equal to 60 and less than 75 then the grade if first division.
- If aggregate is greater than or equal 50 and less than 60, then the grade is second division.
- If aggregate is greater than or equal 40 and less than 50, then the grade is third division.

Code :

```python
marks = []

for i in range(1, 6):
    m = int(input("Enter marks of subject " + str(i) + ": "))
    marks.append(m)
result = "Pass"
for m in marks:
    if m < 40:
        result = "Fail"
        break
total = sum(marks)
percentage = total / 5

print("Total Marks:", total)
print("Percentage:", percentage)

if result == "Fail":
    print("Result: Fail")
else:
    print("Result: Pass")
    if percentage > 75:
        print("Grade: Distinction")
    elif percentage >= 60:
        print("Grade: First Division")
    elif percentage >= 50:
        print("Grade: Second Division")
    elif percentage >= 40:
        print("Grade: Third Division")
```

```
Enter marks of subject 1: 65
Enter marks of subject 2: 56
Enter marks of subject 3: 56
Enter marks of subject 4: 78
Enter marks of subject 5: 87
Total Marks: 342
Percentage: 68.4
Result: Pass
Grade: First Division

=== Code Execution Successful ===
```

2. Solve the Fibonacci sequence using recursive function in Python.

Code :

```python
def fib(n):
    if n <= 1:
        return n
    return fib(n-1) + fib(n-2)

n = int(input("Enter number: "))
print(fib(n))
```

```
Enter number: 6
8

=== Code Execution Successful ===
```

3. Write a Python program to print different patterns.

Code:

```python
n = int(input("Enter number: "))
for i in range(1, n+1):
    print("*" * i)
```

```
Enter number: 5
*
**
***
****
*****

=== Code Execution Successful ===
```

```python
n = int(input("Enter number: "))
for i in range(n, 0, -1):
    print("*" * i)
```

```
Enter number: 6
******
*****
****
***
**
*

=== Code Execution Successful ===
```

Self Study Assignment: 1

Perform all statistical analysis (Average, Max, Min, Sum) on employee data.

Code :

Programiz
Python Online Compiler

main.py

```python
emp = []
n = int(input("Enter number: "))
for i in range(n):
    x = int(input("Enter salary: "))
    emp.append(x)
print("Sum =", sum(emp))
print("Max =", max(emp))
print("Min =", min(emp))
print("Average =", sum(emp)/n)
```

Output

```
Enter number: 3
Enter salary: 45600
Enter salary: 45900
Enter salary: 68550
Sum = 160050
Max = 68550
Min = 45600
Average = 53350.0

=== Code Execution Successful ===
```